

RESEARCH ARTICLE

Effects of crosslinking density and Lewis acidic sites on conductivity and viscoelasticity of dynamic network electrolytes

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Abstract

Polymers are promising materials for replacing organic liquids as electrolytes, and network architectures allow for the modulus to be tuned pseudo-independently of conductivity. When the crosslinks are dynamic bonds, they offer the additional benefits of recyclability and self-healing in response to damage. Dynamic network electrolytes (DNEs) comprised of precise linker lengths of 2, 3, or 4 repeat units of ethylene oxide and boronic ester junctions were prepared to investigate the roles of dense networks and bond exchange on conductivity and rheological properties. A range of salt concentrations were probed, and longer linker lengths led to consistently higher conductivities even after accounting for difference in the glass transition. In contrast, non-monotonic trends are observed in the salt dependence of viscosity as a function of linker length. The interaction of anions from the salt with boron leads to a drop in the viscosity, and at a critical salt content the networks no longer form a percolated network. From the bulk viscosity, a Walden Plot shows a transition from superionic to subionic behavior with added salt. These structure-property relationships offer key valuable insights for designing sustainable electrolytes.

KEYWORDS

dynamic networks, ionic conductivity, rheology, self-healing

1 | INTRODUCTION

Solid polymer electrolytes (SPEs) have garnered significant interest over the past decades because they offer desirable properties (nonflammable, processable, high modulus) that make them safer alternatives to organic liquids,^{1–5} in particular for the case of Li-ion batteries where dendrites can create internal shorts.^{4,6–8} A major challenge in the development of SPEs is to boost the mechanical properties such as modulus while retaining substantial conductivity. One approach is to use block copolymers with high modulus glassy domains and a soft

conductive domain.^{9–14} A second approach is to use a network polymer architecture where crosslinking provides the mechanical integrity. Even with a shear modulus as low as 10^5 Pa, significantly lower than that of a glass (10^9 Pa), it has been shown that networks can effectively suppress dendrites and lead to longer cycling and safer batteries.¹⁵

Understanding how network architectures simultaneously impact conductivity, modulus, glass transition temperatures, and ion clustering is important for designing electrolytes. Studies on permanent networks have shown that mechanically robust and conductive

materials can be achieved through control of crosslink density, salt content, and in some cases the addition of plasticizer.^{16–23} One route to build upon permanent network electrolytes is the incorporation of dynamic bonds into the backbone or at crosslink points. Such dynamic network electrolytes (DNEs) have been shown to be self-healing, dissolvable back to monomer, and stimuli-responsive.^{17,24,25} Bond exchange can also provide an additional dynamic process in the materials which may impact ion transport if the timescales are comparable. A complex interplay between salt addition, dynamic bond kinetics, temperatures, and crosslink density allows for a large degree of tunability in conductivity and rheological properties of DNEs.

Boronic esters are well-known dynamic bonds, and the boron can act as a Lewis acid (electron acceptor) due to an empty p-orbital.^{26–30} In our prior work, the addition of salt to a boronic ester network with ethylene oxide linkers resulted in a large drop in viscosity and modulus as the anion-boron coordination disrupted the network architecture.¹⁷ Boron sites have also been pursued in linear, star, and network polymer electrolytes as they lead to improved salt dissociation and Li mobility by trapping or coordinating with the anion.^{31–36} Fujinami and coworkers first demonstrated that boroxine rings significantly improved the transference number of Li in PEO-based SPEs, reaching values close to unity.³¹ Wakihara and coworkers synthesized network and star PEO-based SPEs with boronic ester junctions,^{32,34} while Ohno and coworkers synthesized linear PEO SPEs with mesitylborane groups.^{33,35} These systems also showed improved Li mobility and conductivities reached 10^{-3} S/cm^{32,34} and 10^{-5} S/cm^{33,35} respectively. Lee and coworkers synthesized gel-polymer electrolytes based on PVDF crosslinked by trigonal boronic esters, which exhibited conductivities up to 10^{-3} S/cm and allowed a battery to retain 87% of its initial capacity over 100 cycles.³⁶ However, most of these previous reports involve electrolytes swollen with small molecules,^{32–36} and none of them commented on the dynamic nature of the B-O bond. Simulation studies have been conducted on systems with Lewis acidic sites and found an enhancement in cation diffusion, further supporting the importance of these sites in polymer electrolytes.^{37,38} Quantifying changes in viscoelasticity with salt addition is critical to a comprehensive picture of DNEs, where both modulus and viscosity are greatly impacted in boronic ester networks.

Our previous study on DNEs investigated one linker length and the role of added salt on bulk relaxation times, modulus, and conductivity.¹⁷ Here, we investigate a systematic series of DNEs using precise linkers with 2, 3, or 4 repeat units of ethylene glycol (2EG, 3EG, 4EG)

crosslinked by boronic esters to probe the role of network confinement on DNEs. Impedance spectroscopy showed that conductivity decreased with shorter linker lengths as the T_g increased. However, after T_g -normalization the 4EG networks still showed the highest conductivity. The content of lithium bistrifluorosulfonimide (LiTFSI) salt in each network was systematically varied, and the conductivity was probed as a function of $r = [\text{Li}]:[\text{EO}]$. Rheological measurements showed the viscosity decreased with added salt, and the magnitude of the drop showed non-monotonic trends with linker length. An additional parameter, $s = [\text{TFSI}]:[\text{B}]$, was also investigated to understand how boron-anion coordination impacts conductivity as well as changes in viscosity and modulus. These DNEs showed a transition from super-ionic to subionic behavior on a Walden plot of molar conductance versus fluidity. These observations highlight the critical role of dynamic bonds on ion transport and mechanical properties needed for the design of next generation solid electrolytes.

2 | EXPERIMENTAL

2.1 | Materials

Anhydrous diethylene glycol (2EG), anhydrous triethylene glycol (3EG), boric acid ($\text{B}(\text{OH})_3$), 99% pure tetraethylene glycol (4EG) and LiTFSI were all purchased from Millipore Sigma. 2EG, 3EG, and 4EG were stored in an Ar glovebox that was maintained at <0.5 ppm O_2 and <0.1 ppm H_2O . Activated molecular sieves were added to 4EG to ensure that the monomer was as dry as possible. CR2032 coin cell components (cases, springs, and stainless-steel spacers) were purchased from MTI Corporation. Anhydrous Methanol (MeOH) was purchased from Macron Fine Chemicals. Two mil thick Kapton tape was purchased from Tape Masters. All chemicals and items were used as received unless otherwise stated.

2.2 | DNE synthesis

The short chain ethylene glycols (2EG, 3EG, or 4EG) were mixed together in a Schlenk flask with $\text{B}(\text{OH})_3$, a calculated amount of LiTFSI, and ~ 5 ml of MeOH as solvent. The reaction mixture was then heated at 60°C in an oil bath to dissolve the $\text{B}(\text{OH})_3$ and LiTFSI, yielding a clear homogenous solution. A dynamic N_2 flow was then connected to the Schlenk flask and allowed to continuously flow while heat was applied in incremental steps at 70°C for 2 h, 120°C for 2 h, and 150°C overnight to

form a transparent viscous fluid or malleable solid. The material was then dried in a vacuum oven at 120 °C overnight, followed by rapid sealing with a stopcock and septa, and transferred into a glovebox.

2.3 | ^{11}B solid state NMR (^{11}B -ssNMR)

A Varian Unity Inova 300 MHz spectrometer was used to collect ^{11}B spectra using 5000 total scans in static mode (no spinning), calibrated using NaBH_4 as the reference. Samples were packed in ceramic 4 mm rotors inside of an Ar glovebox. Spectra were then analyzed using MNOVA software to analyze spectra and extract chemical shift and linewidth values.

2.4 | ATR-Fourier transform infrared

IR spectra was collected using a Bruker ALPHA Attenuated Total Reflectance Fourier transform infrared (ATR-FTIR) spectrometer with a platinum-ATR QuickSnap sampling module. All measurements were performed in a glovebox by scanning from 400 cm^{-1} to 4000 cm^{-1} with 16 total scans. Solid samples were clamped down to the diamond ATR crystal to ensure good contact, while liquid samples were left undisturbed.

2.5 | Coin cell preparation

Samples were prepared in CR2032 coin cells inside of an Ar glovebox. Samples were prepared by pressing the networks using a 1 kg weight at 90 °C between stainless steel discs with Kapton tape as a spacer to yield thicknesses between 200 to 300 μm and diameter of 10 mm. This process ensures that the any thickness changes in the ensuing measurements are minimal. Coin cell components were then assembled in the following order: bottom case/ spring/ disc/ samples/ disc/ spring/ top case. An MTI Corp. MSK-160E crimper was used to assemble the coin cell at a pressure of ~ 9 psi.

2.6 | Differential scanning calorimetry

Samples weighed between 5–8 mg and were sealed in Tzero aluminum pans inside of a glovebox. A TA Instruments Q 20 was used to take all measurements. Experiments were conducted using a heat/cool/heat profile in a temperature range of -80 to 150 °C at a rate of 10 °C /min. The glass transition temperature (T_g) for each sample was obtained from analysis of the second heating

profile using the $\frac{1}{2} \Delta C_p$ criterion in Universal Analysis software (Figures S1-S3).

2.7 | Wide-angle X-ray scattering

Measurements were taken using a Xenocs GeniX3D Cu $K\alpha$ X-ray source (1.54 Å) with a Pilatus 2D detector, where AgBe powder was used to calibrate the sample to detector distance. Networks were packed into 4 mm thick quartz capillary tubes, purchased from Hilgenberg GmbH, inside of a glovebox. The capillaries were sealed using waterproof epoxy to ensure that the networks were not exposed to any water. Spectra were collected under ambient conditions with a 30-minute exposure time per sample. The 2D diffraction plots were obtained using Fit2D software. The normalized plots were constructed by dividing the spectra by their respective maximum amorphous halo peak intensity.

2.8 | Electrochemical impedance spectroscopy

Impedance was measured using a Biologic SP300 potentiostat with a Controlled Environment Sample Holder and Intermediate Temperature System accessories. Impedance spectra were collected by applying an AC potential of 20 mV from high to low frequency (1 MHz to 100 mHz) in 5 °C increments from 90 °C to 10 °C. Only data collected between 90 to 20 °C was used for 2EG due to resolution. The real conductivity (σ') was calculated from complex impedance, $Z^* = Z' + i\omega Z''$ following literature protocol (Figure S4).

$$\sigma'(\omega) = \frac{l}{A} * \frac{Z'(\omega)}{[Z'(\omega) + Z''(\omega)]^2},$$

where l and A are the thickness and area respectively of the network material. All samples were measured using CR2032 coin cells that were prepared following the procedure stated above to ensure an inert environment.³⁹

2.9 | Rheology

Rheological measurements were made on a TA Instruments DHR-2 rheometer fitted with an environmental test chamber accessory using 8 mm diameter parallel plate geometries. Samples were transferred from an Ar glovebox and immediately loaded on to the rheometer pre-heated at 130 °C. The sample was heated at 130 °C

for 2 h to remove any potential H₂O that could be absorbed during sample transfer as well as to eliminate bubbles. The sample was then pressed to a thickness of $\sim 1000\ \mu\text{m}$ on the rheometer and excess material was trimmed. Stress sweeps were performed between 200 to 4000 Pa at 0.5 Hz at 130 °C to probe the linear regime (Figure S5). The temperature was then ramped to 90 °C at a rate of 2 °C/min while applying an appropriate stress at 0.5 Hz. Frequency sweeps were then performed at 90 °C from 100 Hz to 0.1 Hz by applying the same stress chosen for the temperature ramp to extract storage modulus (G'), loss modulus (G''), and complex viscosity (η^*) (Figure S6).

2.10 | Healing demonstration

Networks were pressed into a self-healing apparatus similar to a previously reported configuration.¹⁷ All experiments were conducted inside of a glovebox. The apparatus was placed on top of a hot plate heated at 60 °C and an alternating potential of $\pm 2\ \text{V}$ was applied

using a Biologic SP-200 potentiostat. Damage is introduced by cutting the electrolyte using a razor blade. A 1 kg weight is then placed on top of the electrolyte in order to accelerate healing.

3 | RESULTS AND DISCUSSION

All of the DNEs were synthesized via a condensation reaction using precise ethylene glycols ($n = 2, 3$, or 4) and boric acid with MeOH as the solvent (Figure 1(A)). The LiTFSI was added in controlled amounts, and represented with r , defined as the ratio of [Li]:[EO]. Synthesis was confirmed using both ¹¹B solid-state NMR (ssNMR) and FTIR as shown in Figure 1(B),(C) respectively. The NMR spectra revealed a distinct change for all three networks in the boron site relative to B(OH)₃ which is converted during the condensation. Increasing the linker length resulted in the peak of the boronic ester broadening and shifting downfield.^{40,41} The FTIR spectra showed the disappearance of OH peaks ($\sim 3300\ \text{cm}^{-1}$) and emergence of BO peaks ($\sim 1300\ \text{cm}^{-1}$) after the reaction.⁴² We

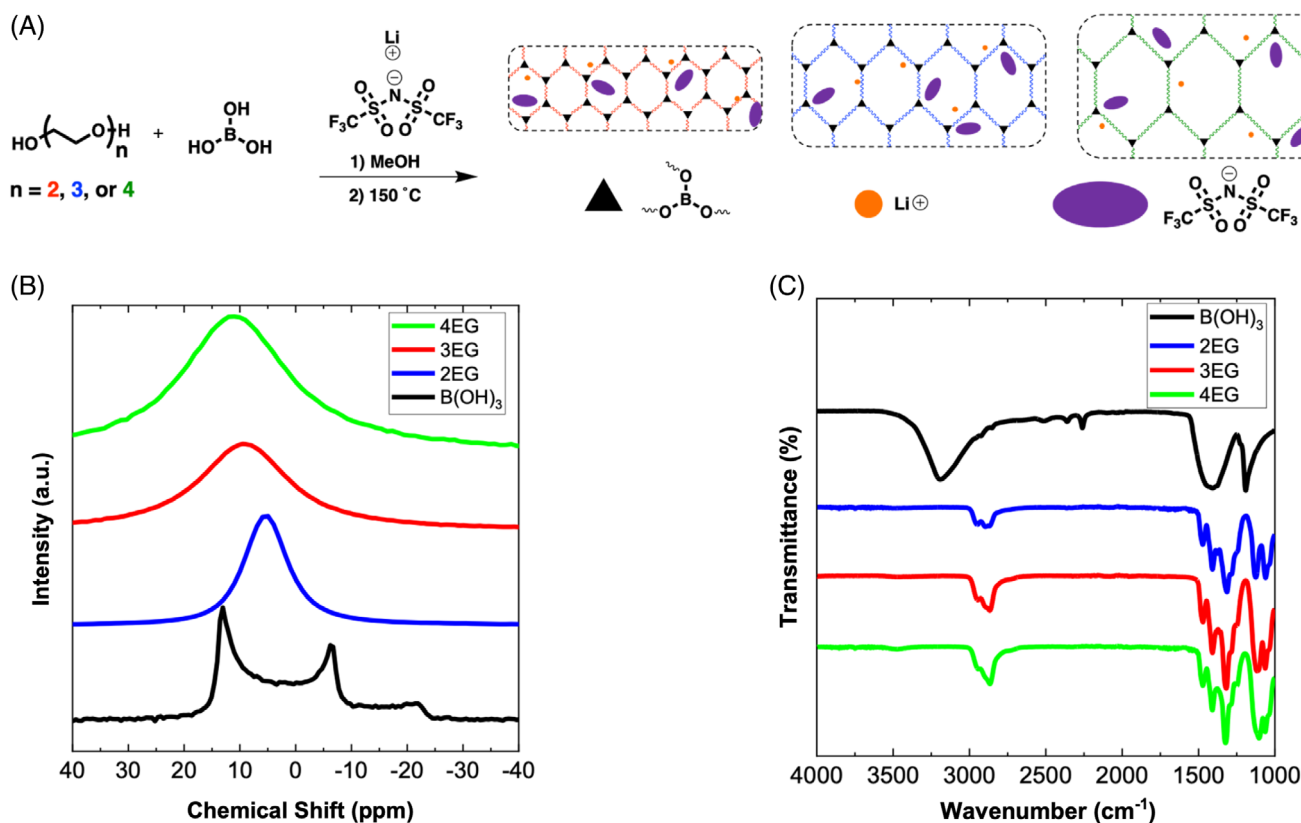


FIGURE 1 (A) Formation of dynamic networks with boronic-esters crosslinks via the condensation reaction of monodisperse oligomeric ethylene glycols and boric acid in the presence of LiTFSI. (B) Boron NMR shows the conversion of B(OH)₃ upon network formation that leads to a new distinct site. (C) ATR-Fourier transform infrared (FTIR) spectra indicate high conversion of network forming reaction through the disappearance of free OH groups and the emergence of a boronic ester peak at $\sim 1300\ \text{cm}^{-1}$ [Color figure can be viewed at wileyonlinelibrary.com]

note that there is still a small signal corresponding to free OH groups which are necessary in order for exchange processes to occur.

The DNEs were then characterized using impedance spectroscopy to measure conductivity. A representative Nyquist plot is shown in Figure 2(A) where the imaginary (Z'') and real (Z') components of impedance are plotted for different temperatures. Frequency dependent conductivity (Figure 2(B)) was then calculated from impedance,^{43,44} and the plateau value is taken as the DC conductivity. A monotonic increase was observed with increasing temperature, which is expected for ionic polymer electrolytes.^{13,18,45–47} Conductivity was then measured as a function of r for the three network linker lengths (Figure 2(c)). Our previous investigation¹⁷

revealed a nonmonotonic trend in conductivity versus r for 3EG networks, where a maximum was observed at $r = 0.085$. This trend is typical in salt-in-polymer systems and arises due to the competition between ion concentration, ion aggregation, and an increase in T_g .^{13,45–48} Initially as salt was added, the T_g and conductivity both increase for all networks. However, at sufficiently high r values the ions begin to aggregate as inferred from wide-angle x-ray scattering (WAXS, Figure S7-S9). As r increases an anion-anion correlation peak appears at $q = 10 \text{ nm}^{-1}$ grows with added salt for all linker lengths as anticipated for salt in polymer systems.^{49–51} With increasing r , the aggregation of ions may lead to a reduction in the mobility or concentration of free ions, leading to a decrease in conductivity. When the linker is shortened to

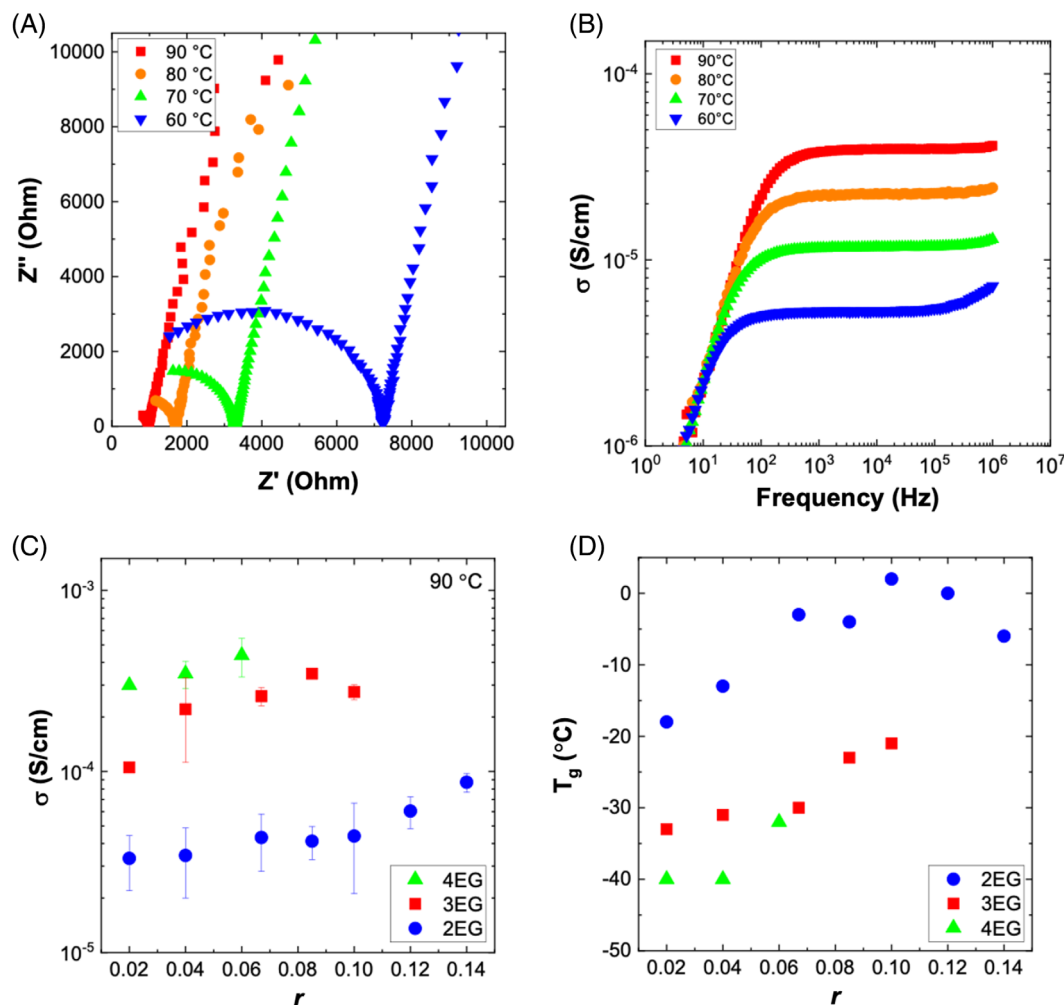


FIGURE 2 (A) A representative Nyquist plot that is constructed from impedance measured from 90°C to 60°C. only four temperature measurements are shown for clarity. (B) Frequency dependent conductivity plots constructed from 1a. A monotonic increase in conductivity is observed with respect to temperature. (C) High temperature conductivities for DNEs with varying linker length plotted as a function of r show that higher crosslinking density leads to a significant decrease in conductivity but allows for higher concentrations of LiTFSI to be incorporated. (D) T_g s for the various DNEs plotted as a function of r , illustrating unique trends for each linker length. Data for 3EG networks with $r = 0.02, 0.067, 0.085$, and 0.1 were taken from reference 17 [Color figure can be viewed at wileyonlinelibrary.com]

2EG, we observe a significant drop in conductivity by a factor of three relative to 3EG networks, while 4EG networks are the most conductive. In contrast to the maximum observed in 3EG networks, the conductivity of 2EG networks show a monotonic increase as r increases from 0.02 to 0.14. Notably, the T_g begins to decrease at $r = 0.1$ in the 2EG networks which leads to the continued increase in conductivity.

In order to determine the role of T_g on differences across linker lengths, the conductivity was normalized by T_g/T as shown for a fixed r value of 0.02 (Figure 3(A)). The data follows a VFT trend, indicating some degree of coupling between ion transport and segmental dynamics.^{44,52–57} Normalizing by T_g leads to conductivities which superimpose except for the 4EG network which is shifted to higher values. We attribute this to the larger network mesh which can provide less hindered transport of ions. When examining networks with a fixed linker length and different salt concentrations, the molar conductivity (Λ) is plotted and normalization by T_g provides superposition of the curves over most of the temperature range indicating that salt is primarily affecting conductivity by altering the segmental dynamics in this range of r values (Figure 3(B)). The superposition of conductivity is also observed in 3EG and 4EG networks (Figure S10 and S11).

Prior work on polymer electrolytes has shown that the incorporation of a Lewis acidic site, for example boron, is beneficial because it can coordinate with the anion and thus increase the fraction of current carried by the cation.^{17,30,32,34,58} To understand the role of boronic esters on conductivity, we analyzed the DNEs with the

parameter $s = [\text{TFSI}]:[\text{B}]$. For a given network, r and s are not independent variables; however, changing the linker length allows for a decoupling of r from s . Plotting conductivity against s (Figure 4) essentially normalizes the salt concentration relative to the boronic ester density of the network. Based on Flory's theory of gelation, the critical conversion for percolated network formation using a trifunctional crosslinker is 0.5.⁵⁹ For the present system, the TFSI anion coordinates with the boronic ester crosslink junction and inhibits the formation of a trigonal crosslink. This effectively reduces the extent of reaction, although not necessarily in a quantitative manner, leading to a breakdown of the network structure. From Figure 4, we observe that DNEs can be synthesized up to $s \sim 0.5$ while still forming a network, but eventually do not percolate with further salt addition. The 4EG networks have the lowest density of boronic esters, and thus require the lowest r value to disrupt the network, while the 2EG networks can tolerate more salt and remain solid-like.

With increasing salt, the DNEs no longer have solid-like behavior, as they easily flow at room temperature. The competition between TFSI anion and OH groups for boron sites is supported by FTIR where an increase in free OH vibration intensity is observed with added salt (Figure 5(A)–(C)). The boron NMR spectra show the peak shifted to lower ppm with increasing r , indicating a change in the local environment and bonding of the boronic ester. However, for 2EG a second peak emerged with added salt at $r = 0.085$, while 3EG and 4EG have a single site for all r values (Figure 5(D)–(F)). Based on a calculation of an all-trans linker conformation, the

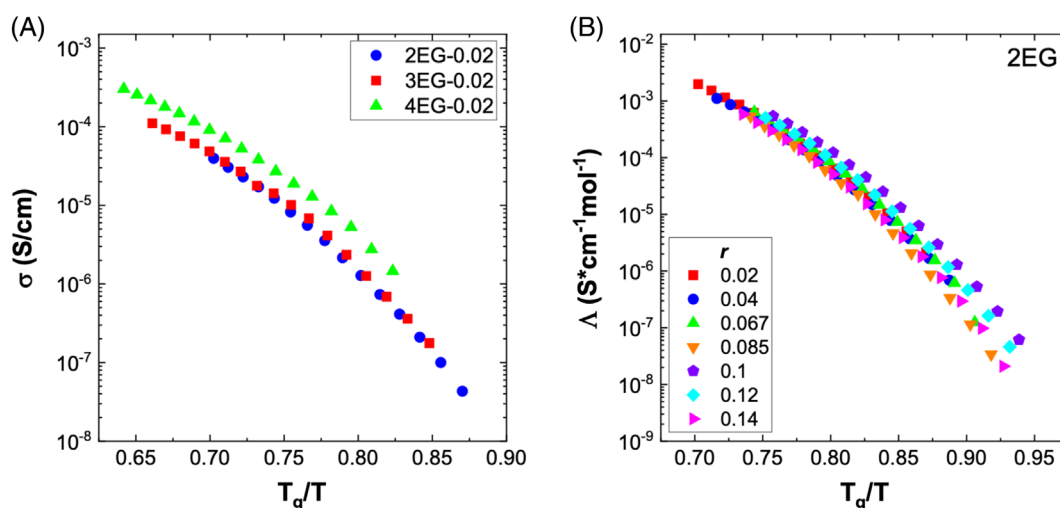


FIGURE 3 (A) Comparison of T_g -normalized-temperature-dependent conductivities for networks with different linker lengths at a fixed LiTFSI concentration ($r = 0.02$). (B) Plot of molar conductivity versus T_g -normalized temperature for 2EG networks across all investigated r values. Both plots highlight that ion transport is strongly coupled to segmental dynamics [Color figure can be viewed at wileyonlinelibrary.com]

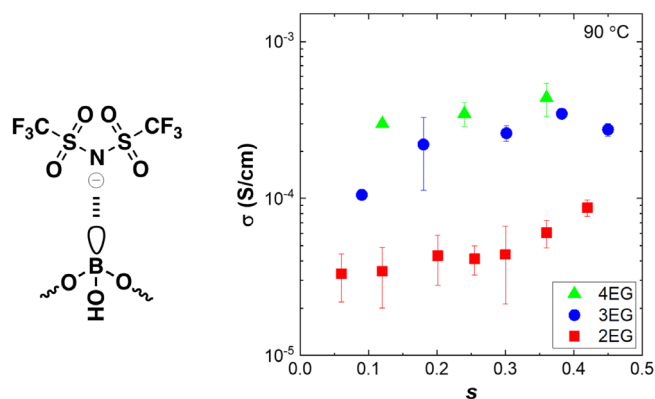


FIGURE 4 The potential structure of the TFSI-B Lewis adduct that forms in the ethylene glycol networks. Conductivity plotted as a function of fraction of B junctions coordinated with TFSI (s), where the trend is similar to that observed when plotted versus r . data for 3EG networks with $r = 0.02, 0.067, 0.085$, and 0.1 were taken from reference 38 [Color figure can be viewed at wileyonlinelibrary.com]

maximum distance between boron sites is 9.5, 13.2, and 16.8 Å for the 2EG, 3EG, and 4EG networks, respectively (Figure S12). The actual spacing will be shorter, and we posit that the anion (TFSI with a long axis of 8 Å)⁶⁰ can span two boron sites simultaneously in the 2EG network. It has been previously shown in small molecule systems that boron can act as a bidentate anion receptor, supporting this claim.^{26,30,61}

Oscillatory shear rheological measurements were conducted to quantify the impact of salt and linker length on the viscoelastic behavior of the DNEs. Frequency sweeps were conducted at 90 °C and a characteristic relaxation time, τ , was defined as the inverse of the crossover frequency where storage modulus (G') is equal to loss modulus (G''). With increasing r values at a fixed linker length (Figure 6(A)), a decrease in rubbery plateau modulus and τ was observed as the anion coordinates with boron. The former is attributed to a decrease in effective crosslink

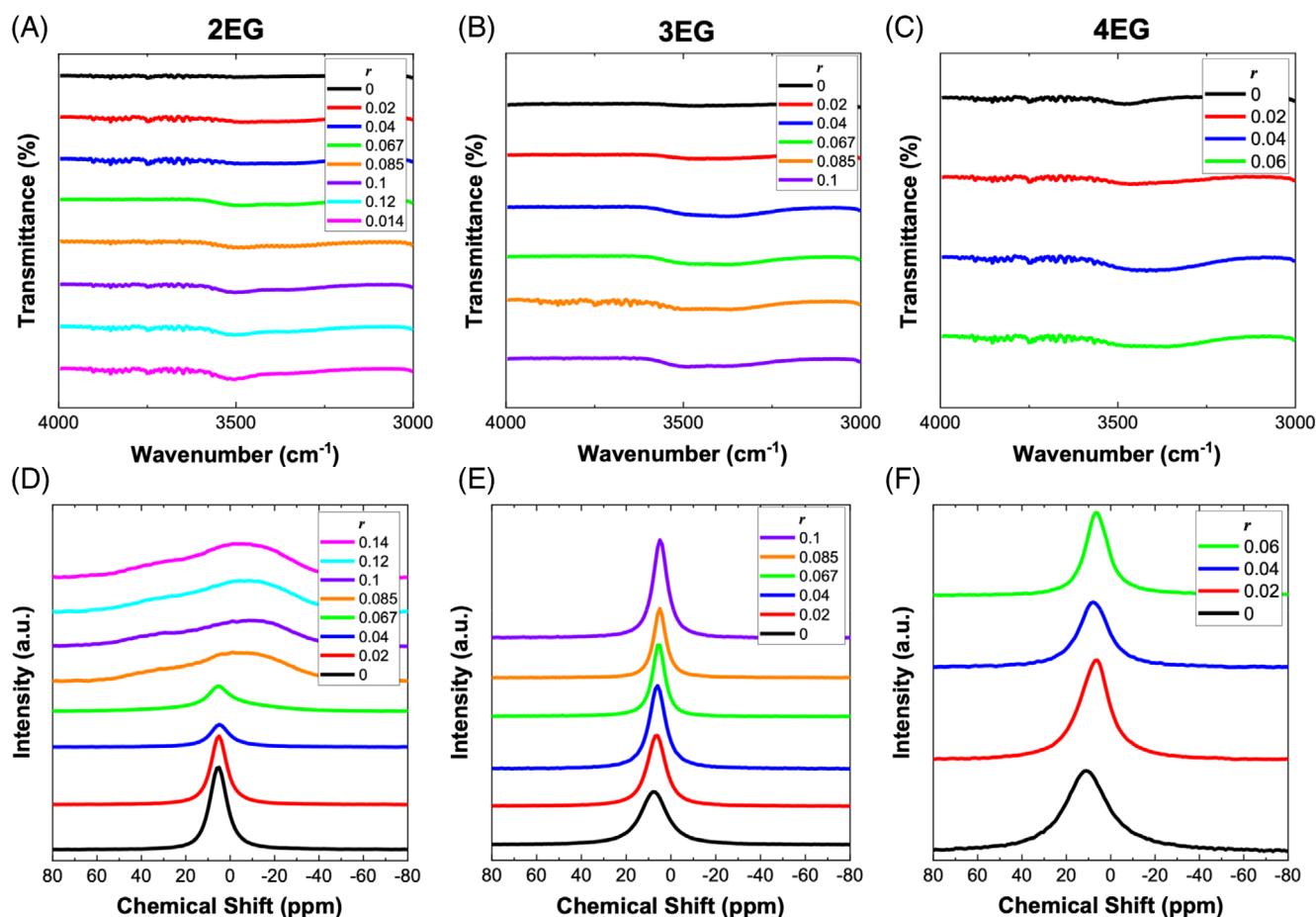


FIGURE 5 Fourier transform infrared (FTIR) spectra between 3000 to 4000 cm⁻¹ for 2EG (A), 3EG (B), and 4EG (C) networks at various r values that illustrate the increase in OH vibration intensity as salt is added. Boron NMR spectra of 2EG (D), 3EG (E), and 4EG (F) networks at various r values that show the boron peak shifts upfield with added salt. Two sites emerge in the 2EG system at $r = 0.085$ and higher that is attributed to TFSI interacting with two boron sites simultaneously due to the high boron density of the networks [Color figure can be viewed at wileyonlinelibrary.com]

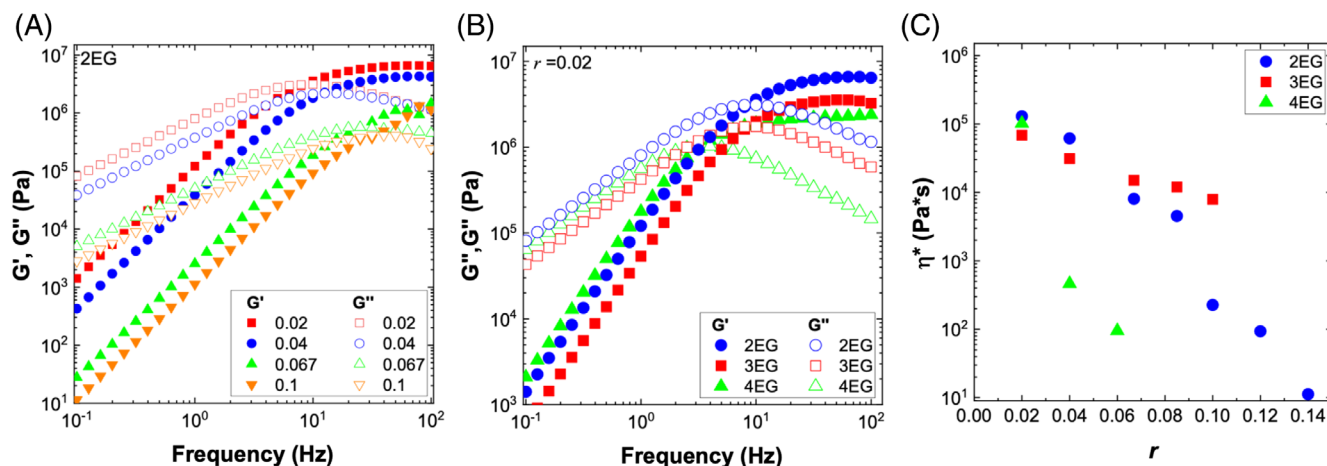


FIGURE 6 (A) Representative frequency sweeps measured at 90°C that show salt addition accelerates relaxation in a 2EG network. (B) Comparison of frequency sweeps across linker lengths at a fixed r value. 2EG and 3EG networks relax on similar timescales, while 4EG is much slower. (C) Complex viscosity plotted against r shows that addition of LiTFSI significantly reduces η^* for 2EG and 4EG, while 3EG is less dramatic [Color figure can be viewed at wileyonlinelibrary.com]

density, while the latter is attributed to an acceleration of the exchange kinetics.¹⁷ At a fixed r value of 0.02, the linker length was varied, and frequency sweeps show a monotonic increase in modulus with decreasing linker length (Figure 6(B)). Despite the anticipated variation of modulus with crosslink density, the 2EG and 3EG networks have nearly identical relaxation times (118 ms and 116 ms respectively), while 4EG takes more than twice as long to relax (296 ms). To further probe the relaxation times as a function of salt content, a plot of complex viscosity's dependence on r is shown in Figure 6(C). The reported viscosities are taken as the plateau value of frequency-dependent viscosity plots (Figure S6). With increasing r , the viscosity of 4EG drops rapidly. The 4EG has the lowest crosslink density initially and thus less salt is required to disrupt the network (Figure 5). The 2EG and 3EG networks have very similar slopes, although the 2EG appears steeper. The 2EG has a denser network, which can allow for a single TFSI to interact with two boron sites simultaneously. Thus, a single anion can disrupt two crosslink junctions and lead to a more significant reduction in crosslink density.

Classical polyether-based electrolytes typically obey the Walden rule, where molar ionic conductivity is controlled by viscosity according to dilute electrolyte theories.⁶² In ionic liquids, the bulk viscosity is used, while for polymer systems, a modified Walden Plot is used with the segmental relaxation time in place of bulk viscosity.^{54,62,63} In the present work, the bulk viscosity is used because the dynamic network flow is controlled by bond exchange events occurring at the network mesh scale. This is supported by the Arrhenius temperature dependence of viscosity in salt free systems with activation

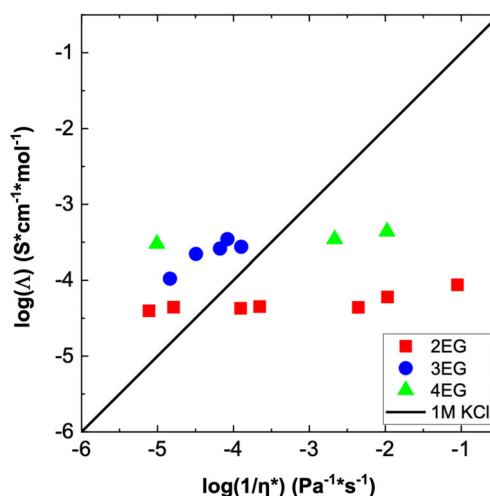


FIGURE 7 Dynamic network electrolytes plotted on a Walden plot illustrating that they do not follow the 1M KCl line. At low r values, 2EG and 4EG networks behave as superionic materials, but transition to subionic as salt is added. The 3EG network is superionic across all r values [Color figure can be viewed at wileyonlinelibrary.com]

energies of 46 kJ/mol (2EG), 41 kJ/mol (3EG) and 26 kJ/mol (4EG), which correspond to boronic ester exchange (Figure S13). Additionally, the network is composed of short strands which have a substantially lower viscosity in the uncrosslinked state. Thus, bond exchange dominated flow is deemed the relevant viscosity for the Walden Plot of DNEs. Figure 7 indicates that adding salt to a given network essentially traverses a horizontal line. The conductivity is minimally affected on a logarithmic scale, while the fluidity (reciprocal viscosity) changes

substantially. At low salt contents, the networks all fall in the superionic regime (above the ideal line). However, 2EG and 4EG transition from superionic to subionic behavior as salt is added, while 3EG acts superionic across all LiTFSI concentrations examined. In the present system, ionic conductivity and bulk viscosity can be tuned essentially independently through control of crosslink density. This observation further highlights that dynamic bonds play a critical role in facilitating ion transport in these materials. As in our prior work,¹⁷ the dynamic bonds also allow for self-healing in response to a cut by a razor blade. The conductivity recovers in response to damage (Figures S14 and S15) indicating that all linker lengths are capable of self-healing. Further studies are needed to provide a systematic investigation of the roles of viscosity, modulus, and salt content on the healing properties of DNEs.

4 | CONCLUSION

Structure–property relationships of DNEs with varying linker length and salt content were reported. The length of the ethylene glycol linker ranged from 2, 3, and 4 repeat units, across a range of r values from 0 to 0.14. Impedance spectroscopy measurements revealed that conductivity increased systematically as the linker length increased. Boron NMR measurements revealed that addition of LiTFSI to the networks resulted in coordination between the boron crosslink junction and TFSI anion. For the 2EG network, it was suggested that at high r values TFSI could interact with two boron sites simultaneously based on broad peaks relative to the 3EG and 4EG networks. Rheology measurements revealed the complex effects of boron density on the DNEs' viscoelastic behavior. Increasing r led to a decrease in modulus and relaxation time due to boron-TFSI adducts disrupting the network topology. However, when comparing the viscosity dependence on r across the linker lengths, the viscosities of 2EG and 4EG networks drop dramatically by 10^3 or 10^4 Pa s, respectively, while the decrease in 3EG viscosity is less severe. This is attributed to the difference in boron density, where less TFSI is needed to disrupt the boron sites in 4EG (lowest density) while salt interacts with two boron sites in 2EG (highest density). On a Walden Plot of molar conductivity versus fluidity, the DNEs all behave superionically at low r , while increasing r caused 2EG and 4EG networks to transition from superionic to subionic behavior, while 3EG remained superionic for all r values. Our findings provide valuable insight on how to tailor linker length influence the properties of DNEs, expanding the design capabilities for next generation Li-transporting electrolytes.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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